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SUMMARY

Assistant Research Professor at the Cambridge Stem Cell Institute leading a research group on epigenetics and ageing; track record of competitive funding and research awards. Full CV and publications available at herzog-lab.com.

KEY TOPICS

-  epigenetics
-  biomarkers & multi-omic human studies
-  ageing research

EDUCATION

2015 – 2019	Ph.D. in Neuroscience (awarded 29 Jun 2019)	University of Edinburgh, UK
2014 – 2015	M.Sc. by Research in Integrative Neuroscience (<i>distinction</i> ★)	University of Edinburgh, UK
2011 – 2014	B.Sc. in Molecular Medicine (<i>distinction</i> ★) graduated top of my class aged 19; joint classes with medical school	Innsbruck Medical University, AT

EXPERIENCE (SELECTED)

2026 – present	Group Leader & Assistant Research Professor • Leading research group on epigenetic ageing and stem cells at the Cambridge Stem Cell Institute	University of Cambridge, UK
2024 – 2026	King's Prize Fellow (group leader) • Established research programme on epigenetic ageing and stem cell function	King's College London, UK
2023 – current	Executive Committee • Leading consensus to establish biomarkers of ageing as reliable tools for longevity interventions • Co-chair of the annual Biomarkers of Aging Conference (Harvard Medical School, 2023-ongoing)	Biomarkers of Aging Consortium, US
2020 – 2024	Senior Scientist and Lecturer • Head of computational biology team (up to 4 postdocs and research assistants), specialising in high-dimensional DNA methylation data analyses; leading several projects • Co-PI of a human multi-omic intervention trial (TirolGESUND, NCT05678426), assessing the impact of lifestyle changes on omic biomarkers of ageing and cancer • Teaching and course material development, supervision of staff and students	Universität Innsbruck, AT
2019 – 2020	Associate Business Development Manager • Scientific liaison for research and clinical collaborations at omics-based precision diagnostics company	BioClavis Ltd., Glasgow, UK
2019 – 2019	Postdoctoral researcher • Investigated immune-derived signalling factors involved in neuroprotection using molecular and computational methods (gene expression analysis, <i>in vivo</i> CRISPR/Cas9, microscopy, data analysis)	University of Edinburgh, UK
2015 – 2019	Ph.D. researcher • Explored the immune system's contribution to central nervous system repair in a regenerative organism (<i>D. rerio</i>) using molecular, cellular, and computational tools (<i>in vivo</i> confocal timelapse microscopy, RNA-seq/qRT-PCR, pharmacological and genetic perturbations, computation analysis)	University of Edinburgh, UK
2014 – 2015	M.Sc. researcher • Extensive biochemical analysis of evolutionary conservation of synaptic protein complexes	University of Edinburgh, UK

PUBLICATIONS (SELECTED)

A full list of >45 peer-reviewed articles can be found on [Google Scholar](#) and [my website](#).
* denotes joint first authorship,  denotes corresponding authorship.

2026	Immune Aging Biomarkers for Clinical Trials Cipriano A, Justice J, [...], & Herzog C  . (10.1038/s41591-026-04493-5)	Nature Medicine
2025	Multi-modal atlas of lifestyle interventions reveals malleability of ageing-linked molecular features bioRxiv Herzog C  , Vavourakis CD, Redl E, [...] & Widschwendter M. (10.1101/2025.08.30.673115)	
2025	Systems epigenetic approach towards non-invasive breast cancer detection Herzog C , Theeuwes B, Jones A, [...] & Widschwendter M. (10.1038/s41467-024-53696-2)	Nature Communications

2025	Functionally enriched epigenetic clocks reveal tissue-specific discordant aging patterns in individuals with cancer Herzog C, Redl E, Barret J, Aminzadeh-Gohari S, [...] & Widschwendter M. (10.1038/s43856-025-00739-4)	Communications Medicine
2024	Challenges and Recommendations for The Translation of Biomarkers of Aging Herzog C*, Goeminne L*, Poganik JR*, [...], & Gladyshev VN. (10.1038/s43587-024-00683-3)	Nature Aging
2024	Cigarette Smoking and E-cigarette Use Induce Shared DNA Methylation Changes Linked to Carcinogenesis Herzog C, [...], & Widschwendter M. (10.1158/0008-5472.CAN-23-2957)	Cancer Research
2024	Validation of Biomarkers of Aging Moqri M*, <u>Herzog C</u> , Poganik JR*, [...], & Gladyshev VN. (10.1038/s41591-023-02784-9)	Nature Medicine
2023	Biomarkers of Aging for the Identification of Longevity Interventions Moqri M*, <u>Herzog C</u> , Poganik JR*, [...], & Gladyshev VN. (10.1016/j.cell.2023.08.003)	Cell
2023	Performance of the WID-qEC test versus sonography to detect uterine cancers in women with abnormal uterine bleeding (EPI-SURE) Evans I, [...], <u>Herzog C</u> , Jurkovic D, & Widschwendter M. (10.1016/S1470-2045(23)00466-7)	Lancet Oncology
2022	A Simple Cervicovaginal Epigenetic Test for Screening and Rapid Triage of Women With Suspected Endometrial Cancer: Validation in Several Cohort and Case/Control Sets Herzog C*, Marin F*, [...] & Widschwendter M. (10.1200/JCO.22.00266)	Journal of Clinical Oncology
2022	The WID-BC-index identifies women with primary poor prognostic breast cancer based on DNA methylation in cervical samples Barrett J*, <u>Herzog C</u> *, [...] & Widschwendter M. (10.1038/s41467-021-27918-w)	Nature Communications

AWARDS

2026	Early Career Award	AACR (American Association for Cancer Research), Cancer Research, US
2025	Rising Star	Longevity Summit Dublin, IE
2024	Prof. Anthony Mellows Medal	Prof. Anthony and Elizabeth Mellows Trust, UK
2024	Life Sciences Award - Basic Science • most prestigious award for outstanding research in the life sciences for early career scientists in Austria	ÖGMBT (Austrian Association of Molecular Life Sciences and Biotechnology), AT
2024	Young Scientist Award	Austrian Society for Gynaecologic Oncology, AT
2024	Prof. Ernst Brandl Prize	Ernst Brandl Stiftung, AT
2022	Falling Walls Lab Winner and Global Finalist	Falling Walls Austria, AT
2018	Runner-up Student Publication of the Year	Scottish Newspaper Society, UK

FUNDING - AWARDED

Secured over £8.5M funding to date.

2026	epiPRIME: Sustained Viral Resilience; £5.8M	Advanced Research + Invention Agency, UK
2026	Wellcome Career Development Award - unlock:epi; £2.8M	Wellcome Trust, UK
2026	Mercator Fellowship, TauAge Project • collaboration with researchers at TU Munich; this fellowship enables outstanding international researchers to enter long-term collaborations with German institutions	DFG, DE
2026	CARICE Seed Funding (co-PI; PI Dr Rebecca Sumner)	KCL, UK
2025	University Partners Networking Award; £8k	Francis Crick Institute, UK
2025	Collaborative Seed Funding Grant; £8.5k	King's Mechanobiology Centre, UK
2024	King's Prize & Prof. Anthony Mellows Medal Fellowship	King's College London, UK
2015	Competitive Ph.D. Scholarship	University of Edinburgh, UK
2016	Travelling Fellowship for two-month exchange to LMU Munich	Company of Biologists, UK

PATENTS

2024	Endometrial cancer detection
2022	Methods for detecting and predicting breast cancer

INVITED TALKS AND KEYNOTES – CONFERENCES (SELECTED)

2026	Biomarkers of Aging Consortium: Updates and Future Directions	GRC Systems Aging, US
2025	Validation of Ageing Biomarkers	EHA-SWG Scientific Workshop, ES
2025	What the epigenome remembers: past, present, and future	Longevity Summit Dublin, IE
2024	The Path to Biomarker Standardization (Keynote)	Foresight Institute Vision Weekend, DE
2024	Biomarkers of Aging Consortium: Establishing Reliable Biomarkers of Aging	GRC Systems Aging, ES
2022	Discordance in general & epithelial ageing signatures	CSHL Mechanisms of Aging
2022	Harnessing the epigenetic footprint for cancer detection and prediction	Young European Scientist Meeting, PT
• I was an invited keynote speaker presenting alongside highly esteemed researchers, such as Nobel Prize winners Prof. Drew Weissman and Prof. William Kaelin		

INVITED TALKS – OUTREACH AND POLICY (SELECTED)

2024	Epigenetics and ageing biomarkers	Long Night of Museums, Futurium Berlin, DE
2024	Panelist at 'A shared definition of longevity medicine in Italy' session	Milan Longevity Summit, IT
2024	Biomarkers of Aging for the Identification of Longevity Interventions	Italian Government Briefing, Rome, IT

PRESS & MEDIA (SELECTED)

2024	Epigenetic impact of smoking and vaping (live TV interview)	Sky News
2024	Vaping 'linked to cancer and damages body like smoking'	Front page article in The Times
2023	Womb cancer 2023 test breakthrough 'cuts diagnosis time'	The Standard Tech & Science Daily Podcast

MENTORSHIP AND TEACHING (SELECTED)

2026 – present	MPhil and BSc courses	University of Cambridge
2023 – 2024	Cell cycle and tumour biology	B.Sc. Biology – Universität Innsbruck, AT
• Development of new materials and lecture delivery		
2022 – 2024	Epigenetics and cancer	M.Sc. Molecular and Cellular Developmental Biology – Universität Innsbruck, AT
• Developed new month-long module on translational epigenetics cancer research from bench to bedside, including personalised prevention, profiling tools, clinical study design, biomarkers, biobanking, and ethics		

CONFERENCE ORGANISATION

2028	GRC Systems Aging (Vice-Chair)	Castelldefels, ES
2023 – current	Biomarkers of Aging Conference	Harvard Medical School, US
2026	Royal Society Meeting on Healthy Ageing	Newcastle, UK
2025	Inaugural Symposium at the Francis Crick Institute	London Ageing Research Network, UK

BOARD AND EDITORIAL ROLES

2025 – 2026	Chair and Founder	London Ageing Research Network, UK
• Established cross-institutional network, raising >£15k in funding for event and seed-funding grant scheme; organised inaugural London Ageing Research Symposium at the Francis Crick Institute (November 2025)		
2025 – current	Associate Editor	Longevity, UK
2025 – current	Early Career Editor	Epigenomics, UK

PUBLIC ENGAGEMENT & SCIENCE COMMUNICATION

2021 – 2022	Pint of Science City Coordinator	Pint of Science Innsbruck, AT
2016 – 2019	President and Editor	Edinburgh University Science Magazine
2016 – 2017	Pint of Science Team Lead	Pint of Science Edinburgh, UK